

ESP32 WiFi Smart Lighting System

MicroPython • ESP32 • WiFi • Smart Home • IoT

WiFi Controlled Multi-Zone Smart Lighting System

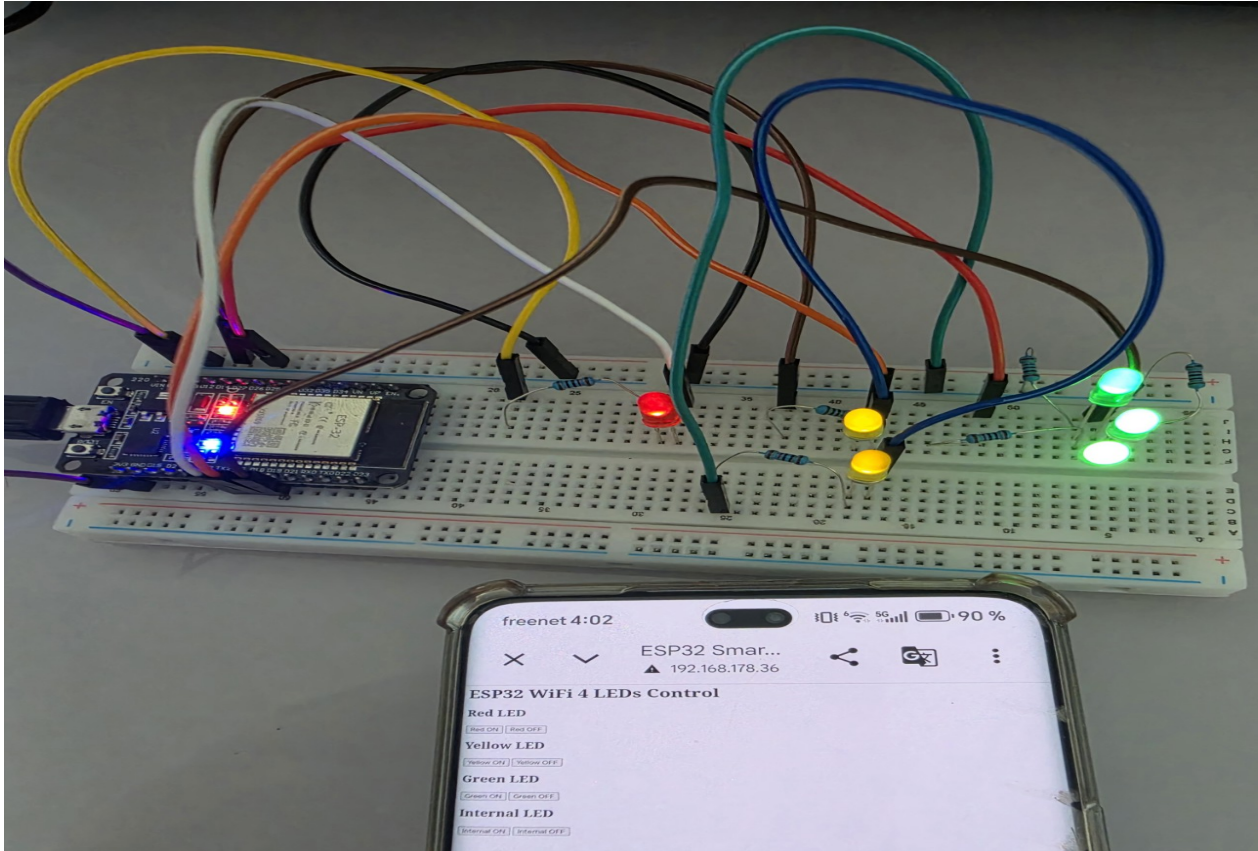


Figure 1: ESP32 WiFi Smart Lighting System

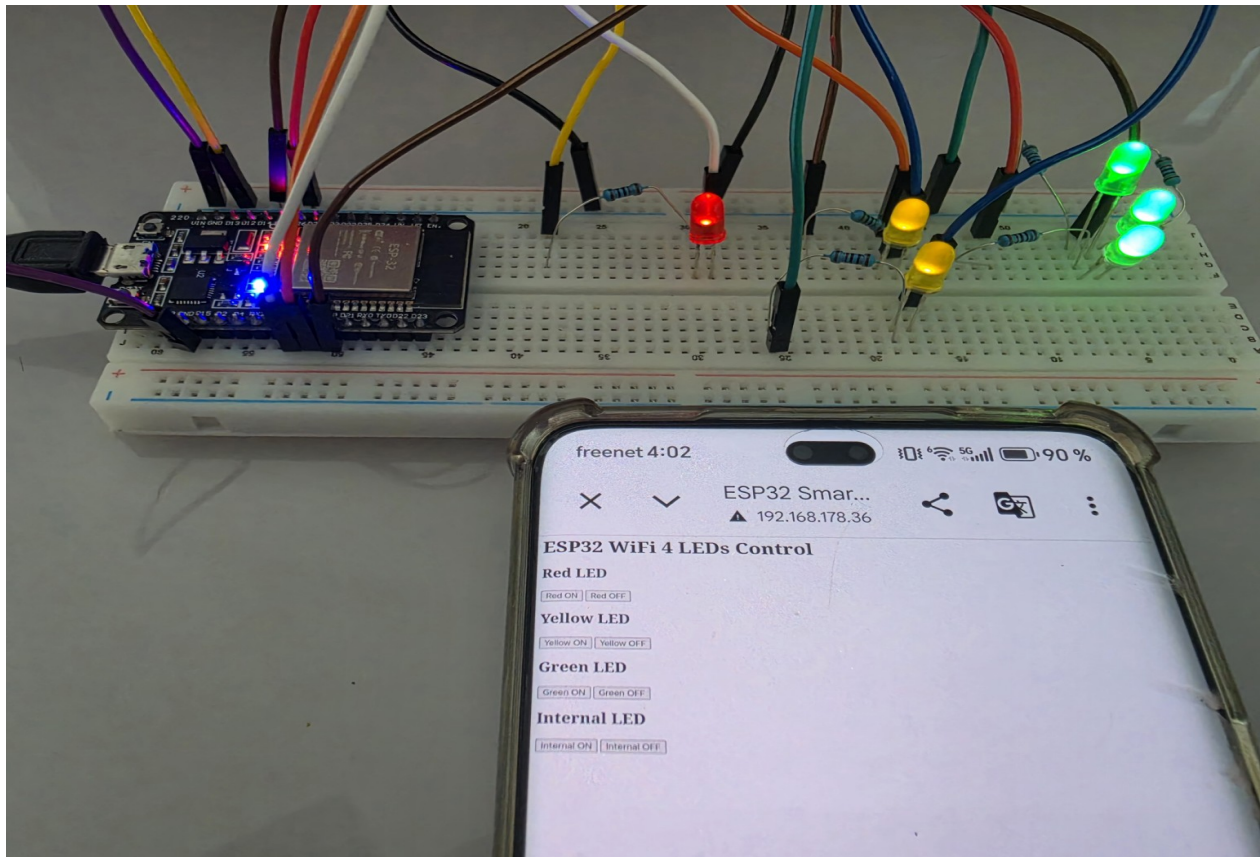


Figure 2: ESP32 WiFi Multi LED Control

■ ■ Deutsche Dokumentation

Dieses Projekt demonstriert ein Smart Home Beleuchtungssystem mit ESP32 und MicroPython. Der ESP32 verbindet sich mit dem WLAN und erstellt einen lokalen Webserver. Über das Smartphone können mehrere LEDs gleichzeitig über den Browser gesteuert werden.

Projektübersicht

Das System enthält mehrere LED-Gruppen: 1 rote LED 2 gelbe LEDs 3 grüne LEDs 1 interne blaue ESP32 LED Alle LEDs werden über WiFi gesteuert.

Verwendete Komponenten

Komponente	Beschreibung
ESP32 DevKit V1	WiFi Mikrocontroller
Rote LED	Lichtzone 1
Gelbe LEDs	Lichtzone 2
Grüne LEDs	Lichtzone 3
Widerstände	LED Schutz
Breadboard	Schaltungsaufbau
Jumper Kabel	Elektrische Verbindungen
MicroPython	Programmiersprache

Parallel LED Verbindung

Mehrere LEDs können parallel an einem GPIO angeschlossen werden. Jede LED besitzt einen eigenen Widerstand für eine sichere Strombegrenzung.

Python Code

```
import network
import socket
from machine import Pin

ssid = 'Your_WiFi_Name'
password = 'Your_WiFi_Password'

red_led = Pin(5, Pin.OUT)
yellow_led = Pin(18, Pin.OUT)
green_led = Pin(19, Pin.OUT)
internal_led = Pin(2, Pin.OUT)

wifi = network.WLAN(network.STA_IF)
wifi.active(True)
wifi.connect(ssid, password)

while not wifi.isconnected():
    pass

addr = socket.getaddrinfo('0.0.0.0', 80)[0][-1]

server = socket.socket()
server.bind(addr)
server.listen(1)

while True:
    client, addr = server.accept()
```

```
request = client.recv(1024)
request = str(request)

if '/red_on' in request:
    red_led.on()

if '/red_off' in request:
    red_led.off()

if '/yellow_on' in request:
    yellow_led.on()

if '/yellow_off' in request:
    yellow_led.off()

if '/green_on' in request:
    green_led.on()

if '/green_off' in request:
    green_led.off()

if '/internal_on' in request:
    internal_led.on()

if '/internal_off' in request:
    internal_led.off()

client.close()
```

English Documentation

This project demonstrates a Smart Home Lighting System using ESP32 and MicroPython. The ESP32 connects to WiFi and creates a local web server. Using a smartphone browser, multiple LED groups can be controlled remotely.

Project Overview

The system includes: 1 Red LED 2 Yellow LEDs 3 Green LEDs 1 Internal ESP32 Blue LED All LEDs are controlled remotely over WiFi.

Components Used

Component	Description
ESP32 DevKit V1	WiFi microcontroller
Red LED	Lighting zone 1
Yellow LEDs	Lighting zone 2
Green LEDs	Lighting zone 3
Resistors	LED protection
Breadboard	Circuit prototyping
Jumper Wires	Electrical connections
MicroPython	Programming language

Parallel LED Connection

Multiple LEDs can be connected in parallel to one GPIO pin. Each LED uses its own resistor for safe current limiting and stable operation.

Python Code

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import network
import socket
from machine import Pin

ssid = 'Your_WiFi_Name'
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red_led = Pin(5, Pin.OUT)
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addr = socket.getaddrinfo('0.0.0.0', 80)[0][-1]

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request = client.recv(1024)
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if '/red_on' in request:
    red_led.on()

if '/red_off' in request:
    red_led.off()

if '/yellow_on' in request:
    yellow_led.on()

if '/yellow_off' in request:
    yellow_led.off()

if '/green_on' in request:
    green_led.on()

if '/green_off' in request:
    green_led.off()

if '/internal_on' in request:
    internal_led.on()

if '/internal_off' in request:
    internal_led.off()

client.close()
```

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